Reliability, Measurement Error, and Score Precision

PSYCHOMETRICIAN · psych-assessment-research-statistics · BoardPrep lesson v1

Official examination scope

The PRC Psychology board page lists Resolution No. 02 s. 2023 and its enhanced TOS for the Psychometrician examination. This lesson uses testing standards and statistical instruction to explain reliability and score precision. Its formula example is a classical-test-theory model, not a universal threshold or frequency claim.

Learning objectives

- Distinguish reliability evidence from validity evidence and a guarantee about an individual
- score.
- Compute a standard error of measurement from a stated SD and reliability estimate.
- Interpret score consistency for the relevant population, administration, and use.

1. Reliability is contextual

Reliability evidence concerns score consistency or precision for a defined population and use. Estimates may address different error sources, such as internal consistency, alternate forms, or occasions. The value can change with examinee range, item set, administration, and scoring. Report the source and context instead of treating reliability as a permanent property of a test.

2. Reliability does not establish validity

Validity concerns whether evidence supports the proposed score interpretation and use. Consistent scores may still represent the wrong construct, omit important content, or support an unjustified decision. An internal-consistency coefficient alone does not show adequate content coverage, fairness, or validity for a particular use.

3. Estimate precision with SEM

Under a classical-test-theory model, SEM = SD_x x sqrt(1 - r_xx), where SD_x is the observed-score standard deviation and r_xx is the relevant reliability estimate. Inputs must refer to an appropriate group and model. SEM summarizes score imprecision; it is not the standard error of a sample mean.

4. Match evidence to a decision

Use reliability evidence relevant to the present population, administration, and decision. A small or unrelated sample may provide weak guidance. For scores near a decision threshold, acknowledge uncertainty and consider additional evidence instead of treating one observed score as error-free.

Compute a classical SEM estimate

Question: SD_x = 10 points and a relevant reliability estimate r_xx = .84. Estimate SEM.
1. Compute 1 - .84 = .16.
2. Take sqrt(.16) = .40.
3. Multiply 10 points x .40 = 4 points.
Answer: Estimated SEM = 4 score points under the stated model and inputs.
Check: The output has the score's unit; it is not a validity coefficient or a guaranteed interval for each person.

Common misconceptions

- A high internal-consistency coefficient proves that the test measures the intended construct.

- One reliability estimate transfers unchanged to every population and decision.
- SEM is the standard error of a sample mean or a guaranteed error bound for one person.
- More similar items always improve construct representation.

Practice patterns

- Compute SEM from the supplied reliability estimate and score SD.
- Explain why evidence from another population may not support the current decision.
- Separate internal consistency from validity and content representation.

Official scope references

Official scope: Enhanced Tables of Specifications for the Psychologists and Psychometricians Licensure Examinations — Professional Regulation Commission; Professional Regulatory Board of Psychology — https://prc.gov.ph/Pages/PRBv4/Psychology.htm

Instructional references

Instructional reference: Standards for Educational and Psychological Testing — AERA, APA, and NCME — https://www.apa.org/science/programs/testing/standards
Instructional reference: Introductory Statistics 2e — OpenStax; Illowsky and Dean — https://openstax.org/books/introductory-statistics-2e/pages/index

BoardPrep lesson material is original instruction. It is not an official examination document.
No claim about question frequency or examination weighting is made.